ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MATHEMATICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) fundamental subject, Level 4. Functions and inverses (15), differential calculus (35), financial mathematics – annuities and loans (20), trigonometry (15), analytical geometry and statistics (15).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 4. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number the answers according to the numbering system used in this question paper.
3. Show ALL calculations, diagrams and graphs that you used to find your answers. Answers only will not necessarily earn full marks.
4. An approved scientific calculator (non-programmable and non-graphical) may be used, unless stated otherwise.
5. Round off answers to TWO decimal places, unless stated otherwise.
6. Diagrams are not necessarily drawn to scale.
7. Write neatly and legibly.

QUESTION 1: FUNCTIONS AND INVERSES

1.1 Given $f(x) = 3x - 6$:

1.1.1 Determine $f^{-1}(x)$. (2)

1.1.2 Calculate $f^{-1}(3)$. (1)

1.2 Given $g(x) = 2x^2$, $x \geq 0$:

1.2.1 Determine $g^{-1}(x)$. (3)

1.2.2 Explain why the domain of g has been restricted to $x \geq 0$. (2)

1.3 Sketch the graphs of $y = 3^x$ and its inverse $y = \log_3 x$ on the same set of axes. Show the line $y = x$ and the intercepts with the axes. (4)

1.4 Given $h(x) = \log_3 x$:

1.4.1 Calculate $h(27)$. (1)

1.4.2 For which values of x is $h(x) < 0$? (2)

[15]

QUESTION 2: DIFFERENTIAL CALCULUS

2.1 Evaluate: $\lim_{x \to 3} \dfrac{x^2 - 9}{x - 3}$ (2)

2.2 Determine $f'(x)$ from first principles if $f(x) = 2x^2 - 3$. (5)

2.3 Determine $\dfrac{dy}{dx}$ if:

2.3.1 $y = 4x^3 - \dfrac{2}{x} + \sqrt{x}$ (4)

2.3.2 $y = (2x - 1)^2$ (2)

2.4 Determine the equation of the tangent to the curve $f(x) = x^2 - 4x + 1$ at the point where $x = 3$. (4)

2.5 Given $g(x) = x^3 - 3x^2 - 9x + 5$

2.5.1 Calculate the coordinates of the turning points of g. (5)

2.5.2 Calculate the coordinates of the point of inflection. (2)

2.5.3 Sketch the graph of g, showing the turning points, the point of inflection and the y-intercept. (4)

2.6 A farmer has 120 m of fencing to fence off a rectangular camp next to a straight river. No fence is needed along the river. Let the sides perpendicular to the river be x metres.

2.6.1 Show that the area is $A = 120x - 2x^2$. (2)

2.6.2 Determine the dimensions that give the maximum area, and calculate this area. (3)

2.7 The distance (in metres) travelled by a robot is given by $s(t) = 5t^2 + 2t$, where t is in seconds. Calculate its velocity after 3 seconds. (2)

[35]

QUESTION 3: FINANCIAL MATHEMATICS

3.1 Lerato saves R800 at the end of every month for 5 years in an account that pays 10% per annum, compounded monthly. Calculate the value of her savings after 5 years. (5)

3.2 A car costing R250 000 is financed over 5 years at 12% per annum, compounded monthly. The first payment is made one month after the loan is granted.

3.2.1 Calculate the monthly payment. (4)

3.2.2 Calculate the total interest paid over the 5 years. (2)

3.2.3 Calculate the outstanding balance immediately after the 24th payment. (4)

3.3 Explain the difference between a future-value annuity and a present-value annuity, with ONE example of each. (2)

3.4 Which is better for an investor: 11% per annum compounded monthly, or 11,4% per annum compounded annually? Show your calculations. (3)

[20]

QUESTION 4: TRIGONOMETRY

4.1 Without using a calculator, show that $\cos 75° = \dfrac{\sqrt{6} - \sqrt{2}}{4}$. (4)

4.2 A vertical mast FT is 45 m high. From point P on the level ground, the angle of elevation of the top T is 28°. Q is another point on the level ground such that FQ = 60 m and the angle $P\hat{F}Q = 110°$.

4.2.1 Calculate the distance FP. (3)

4.2.2 Calculate the distance PQ. (4)

4.3 Solve for x if $x \in [0° ; 360°]$: $\cos 2x = \sin x$ (4)

[15]

QUESTION 5: ANALYTICAL GEOMETRY AND STATISTICS

5.1 A circle has its centre at M(2 ; −3) and a radius of 5 units.

5.1.1 Write down the equation of the circle. (2)

5.1.2 Show that the point P(5 ; 1) lies on the circle. (2)

5.1.3 Determine the equation of the tangent to the circle at P. (4)

5.2 The number of faults found on five production lines in one day is: 4 8 6 10 12

5.2.1 Calculate the mean and the standard deviation. (5)

5.2.2 How many of the values lie within one standard deviation of the mean? (2)

[15]

TOTAL: 100

END OF QUESTION PAPER